

AutoBreaking Change Engineering V2

Risk Matrix, Cascading Impact and Safe Refactor Waves

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Comprehensive Engineering Reference

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AutoBreaking V2 Change Engineering Book

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AutoBreaking V2 Change Engineering Book

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Executive Summary

Metric Value
--- ---:
Rust files scanned 60
Functions detected 848
Critical risk functions 66
High risk functions 52
Medium risk functions 310
Low risk functions 420

Visual Roadmap

flowchart LR
W1[Wave 1\nProtocol Contracts] --> W2[Wave 2\nHardware Abstraction]
W2 --> W3[Wave 3\nReport Schema Stabilization]
W3 --> W4[Wave 4\nPerformance and Observability]
W4 --> W5[Wave 5\nCI and Evidence Hardening]

1. Matriz de Risco por Funcao Critica

<details>
<summary>Expand full critical and high risk matrix</summary>

Risco Funcao Local Linha Justificativa Testes Minimos
--- --- --- ---: --- ---
critical handle_uds_serial_request src/bin/cat_comm_bridge.rs 101 impacts diagnostic protocol contract and ECU session state
critical missing_for_live_flash src/io/hw.rs 49 impacts firmware write path, transfer integrity, and production
critical declared_capabilities src/io/hw.rs 231 impacts hardware transport abstraction and live connectivity
critical open_real_adapter src/io/hw.rs 298 impacts hardware transport abstraction and live connectivity can
critical probe_live_adapter src/io/hw.rs 324 impacts hardware transport abstraction and live connectivity ca
critical append_flash_audit src/io/live_runner.rs 484 impacts firmware write path, transfer integrity, and pro
critical run_flash_preflight src/io/live_runner.rs 749 impacts firmware write path, transfer integrity, and p
critical j1939_uds_req_id src/io/live_runner.rs 816 impacts diagnostic protocol contract and ECU session state
critical is_uds_response_id src/io/live_runner.rs 820 impacts diagnostic protocol contract and ECU session sta
critical recv_can_uds_response src/io/live_runner.rs 1046 impacts diagnostic protocol contract and ECU session

```

| critical | send_uds_and_wait | src/io/live_runner.rs | 1164 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo test --features=live_runner
| critical | live_flash_ecm_firmware | src/io/live_runner.rs | 1286 | impacts firmware write path, transfer integrity, and production gate | cargo check ; cargo test --features=live_runner
| critical | uds_response_id_matches_expected_target | src/io/live_runner.rs | 1631 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo test --features=live_runner
| critical | uds_response_id_rejects_wrong_pf_ps_or_sa | src/io/live_runner.rs | 1639 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo test --features=live_runner
| critical | run_all_phases | src/io/production_program.rs | 77 | impacts release gate and conformance reporting | cargo check ; cargo test --features=production_program
| critical | run_sim_identity_check | src/io/production_program.rs | 601 | impacts release gate and conformance reporting | cargo check ; cargo test --features=production_program
| critical | build_conformance_summary | src/io/production_program.rs | 613 | impacts release gate and conformance reporting | cargo check ; cargo test --features=production_program
| critical | run_uds_runtime_self_test | src/io/production_program.rs | 744 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo test --features=production_program
| critical | supports | src/io/vendor_cat_comm.rs | 30 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | open | src/io/vendor_cat_comm.rs | 88 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | ensure_connected | src/io/vendor_cat_comm.rs | 154 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | send_bridge_request | src/io/vendor_cat_comm.rs | 164 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | read_bridge_response | src/io/vendor_cat_comm.rs | 182 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | append_diag_record | src/io/vendor_cat_comm.rs | 201 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | init | src/io/vendor_cat_comm.rs | 233 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | read_frame | src/io/vendor_cat_comm.rs | 246 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | try_read_frame | src/io/vendor_cat_comm.rs | 261 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | send_frame | src/io/vendor_cat_comm.rs | 273 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | close | src/io/vendor_cat_comm.rs | 286 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | adapter_info | src/io/vendor_cat_comm.rs | 305 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | resolve_bridge_exe | src/io/vendor_cat_comm.rs | 318 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | map_bridge_error | src/io/vendor_cat_comm.rs | 347 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | bridge_info_from_response | src/io/vendor_cat_comm.rs | 368 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | ensure_bridge_compatibility | src/io/vendor_cat_comm.rs | 375 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | now_ms | src/io/vendor_cat_comm.rs | 413 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_cat_comm
| critical | engineering_oil_catalog | src/leak_physics.rs | 496 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=leak_physics
| critical | uds_process_sa | src/main.rs | 6236 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo test --features=main
| critical | uds_send_and_log | src/main.rs | 6246 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo test --features=main
| critical | run_uds_flash_pipeline | src/main.rs | 6278 | impacts firmware write path, transfer integrity, and production gate | cargo check ; cargo test --features=main
| critical | tab_uds | src/main.rs | 9040 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo test --features=main
| critical | fmt | src/uds.rs | 42 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | new | src/uds.rs | 237 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | tick | src/uds.rs | 276 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | process | src/uds.rs | 293 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_session_control | src/uds.rs | 320 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_ecu_reset | src/uds.rs | 353 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_clear_dtc | src/uds.rs | 395 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_read_dtc | src/uds.rs | 428 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_read_data_by_id | src/uds.rs | 504 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_security_access | src/uds.rs | 554 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo check --features=uds
| critical | svc_write_data_by_id | src/uds.rs | 640 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_routine_control | src/uds.rs | 684 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | svc_request_download | src/uds.rs | 733 | impacts firmware write path, transfer integrity, and production gate | cargo check ; cargo check --features=uds
| critical | svc_transfer_data | src/uds.rs | 767 | impacts firmware write path, transfer integrity, and production gate | cargo check ; cargo check --features=uds
| critical | svc_request_transfer_exit | src/uds.rs | 809 | impacts firmware write path, transfer integrity, and production gate | cargo check ; cargo check --features=uds
| critical | svc_tester_present | src/uds.rs | 830 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | record_dtc_set | src/uds.rs | 850 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | nrc | src/uds.rs | 879 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | log | src/uds.rs | 883 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | service_name | src/uds.rs | 905 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | dtc_to_3bytes | src/uds.rs | 926 | impacts runtime behavior and downstream callers | cargo check ; cargo check --features=uds
| critical | session_default_relocks_security | src/uds.rs | 939 | impacts diagnostic protocol contract and ECU session state | cargo check ; cargo check --features=uds
| critical | transfer_exit_rejects_incomplete_download | src/uds.rs | 949 | impacts firmware write path, transfer integrity, and production gate | cargo check ; cargo check --features=uds
| critical | bin_path | tests/vendor_bridge_e2e.rs | 9 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_bridge_e2e
| critical | unique_temp_dir | tests/vendor_bridge_e2e.rs | 22 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=vendor_bridge_e2e
| critical | cat_comm_bridge_probe_and_live_phases_flash_e2e | tests/vendor_bridge_e2e.rs | 33 | impacts firmware write path, transfer integrity, and production gate | cargo check ; cargo test --features=vendor_bridge_e2e
| high | from_path | src/io/allowlist.rs | 33 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | is_allowed | src/io/allowlist.rs | 42 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | per_rule_rate_limit | src/io/allowlist.rs | 46 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | validate | src/io/allowlist.rs | 53 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | rule_matches_frame | src/io/allowlist.rs | 95 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | parse_u32_hex_or_dec | src/io/allowlist.rs | 137 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | parse_mask | src/io/allowlist.rs | 146 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | match_hex_pattern | src/io/allowlist.rs | 150 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | can_rule_matches_mask | src/io/allowlist.rs | 182 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | serial_rule_matches_pattern | src/io/allowlist.rs | 209 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=allowlist
| high | write_intent_enabled | src/io/hw.rs | 223 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --features=hw
| high | connect_ecm | src/io/live_runner.rs | 174 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=live_runner
| high | live_preflight_ecm | src/io/live_runner.rs | 793 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=live_runner
| high | check_live_channel_policy | src/io/live_runner.rs | 1216 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=live_runner
| high | on_rate_limited | src/io/metrics.rs | 37 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=metrics
| high | inject_disconnect | src/io/mock.rs | 32 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=mock
| high | clear_disconnect | src/io/mock.rs | 36 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=mock
| high | read_frame | src/io/mock.rs | 61 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=mock
| high | try_read_frame | src/io/mock.rs | 89 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=mock
| high | send_frame | src/io/mock.rs | 114 | impacts runtime behavior and downstream callers | cargo check ; cargo test --features=mock

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| high | default | src/io/production_program.rs | 69 | impacts release gate and conformance reporting | cargo check ; cargo test
| high | run_sim_baseline_check | src/io/production_program.rs | 568 | impacts release gate and conformance reporting | cargo check ; cargo test
| high | run_sim_connectivity_check | src/io/production_program.rs | 587 | impacts release gate and conformance reporting | cargo check ; cargo test
| high | run_sim_preflight_check | src/io/production_program.rs | 731 | impacts release gate and conformance reporting | cargo check ; cargo test
| high | run_sim_stress_check | src/io/production_program.rs | 763 | impacts release gate and conformance reporting | cargo check ; cargo test
| high | hex_sha | src/io/production_program.rs | 784 | impacts release gate and conformance reporting | cargo check ; cargo test
| high | now_ms | src/io/production_program.rs | 792 | impacts release gate and conformance reporting | cargo check ; cargo test
| high | new | src/io/rate_limiter.rs | 13 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | try_take | src/io/rate_limiter.rs | 23 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | refill | src/io/rate_limiter.rs | 33 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | new | src/io/rate_limiter.rs | 49 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | check_can | src/io/rate_limiter.rs | 57 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | check_serial | src/io/rate_limiter.rs | 70 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | limiter_blocks_when_empty | src/io/rate_limiter.rs | 80 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | aggregate_group_report | src/leak_physics.rs | 73 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | export_calibration_report_json | src/leak_physics.rs | 1654 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | export_calibration_report_csv | src/leak_physics.rs | 1670 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | report_captures_pressure_band_and_safe_window | src/leak_physics.rs | 2030 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | calibration_mode_ingests_csv_and_exports_reports | src/leak_physics.rs | 2107 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | export_leak_report_json | src/lib.rs | 1053 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | export_leak_report_csv | src/lib.rs | 1060 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | export_leak_calibration_report_json | src/lib.rs | 1118 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | export_leak_calibration_report_csv | src/lib.rs | 1126 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test
| high | pick_save_path | src/main.rs | 542 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | leak_ascii_cad | src/main.rs | 6360 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | fmt | src/v2x_telematics.rs | 322 | impacts runtime behavior and downstream callers | cargo check ; cargo test
| high | write_test_allowlist | tests/io_mock.rs | 10 | impacts write authorization and operational safety | cargo check ; cargo test
| high | mock_adapter_allows_write_when_policy_satisfied | tests/io_mock.rs | 97 | impacts hardware transport abstraction and live simulation | cargo check ; cargo test
| high | mock_adapter_dry_run_logs_without_physical_tx | tests/io_mock.rs | 125 | impacts hardware transport abstraction and live simulation | cargo check ; cargo test
| high | mock_adapter_rate_limit_blocks_burst | tests/io_mock.rs | 157 | impacts hardware transport abstraction and live simulation | cargo check ; cargo test
| high | allowlist_parse_mask_accepts_hex | tests/io_mock.rs | 189 | impacts write authorization and operational safety | cargo check ; cargo test
| high | e2e_hydraulic_plus_ac_stress_generates_reports | tests/leak_system_integration.rs | 4 | impacts evidence schema consumed by automation and approvals | cargo check ; cargo test

```

</details>

2. Se Mudar X Entao Quebra Y

<details>

<summary>Expand full X->Y impact table with line references</summary>

```

| X alterado | Onde | Risco | Y afetado | Linhas de referencia | Efeito esperado |
|---|---|---|---|---|---|
| handle_uds_serial_request | src/bin/cat_comm_bridge.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/bin/cat_comm_bridge.rs:100 | impactos na integridade dos dados de diagnostico
| missing_for_live_flash | src/io/hw.rs | critica | src/io/production_program.rs ; src/io/production_program.rs ; src/io/live_runner.rs | src/io/production_program.rs:100 ; src/io/production_program.rs:101 ; src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| declared_capabilities | src/io/hw.rs | critica | src/io/production_program.rs | src/io/hw.rs:231 ; src/io/production_program.rs:232 | impactos na integridade dos dados de diagnostico
| open_real_adapter | src/io/hw.rs | critica | src/io/live_runner.rs ; src/io/live_runner.rs ; src/io/live_runner.rs ; src/io/vendor_cat_comm.rs | src/io/live_runner.rs:100 ; src/io/live_runner.rs:101 ; src/io/live_runner.rs:102 ; src/io/vendor_cat_comm.rs:100 | impactos na integridade dos dados de diagnostico
| probe_live_adapter | src/io/hw.rs | critica | src/bin/simulator_cli.rs ; tests/vendor_bridge_e2e.rs | src/bin/simulator_cli.rs:100 ; tests/vendor_bridge_e2e.rs:100 | impactos na integridade dos dados de diagnostico
| append_flash_audit | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| run_flash_preflight | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| j1939_uds_req_id | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| is_uds_response_id | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| recv_can_uds_response | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| send_uds_and_wait | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| live_flash_ecm_firmware | src/io/live_runner.rs | critica | src/io/production_program.rs ; src/main.rs | src/io/production_program.rs:100 ; src/main.rs:100 | impactos na integridade dos dados de diagnostico
| uds_response_id_matches_expected_target | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| uds_response_id_rejects_wrong_pf_ps_or_sa | src/io/live_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/live_runner.rs:100 | impactos na integridade dos dados de diagnostico
| run_all_phases | src/io/production_program.rs | critica | src/bin/simulator_cli.rs ; tests/vendor_bridge_e2e.rs | src/bin/simulator_cli.rs:100 ; tests/vendor_bridge_e2e.rs:100 | impactos na integridade dos dados de diagnostico
| run_sim_identity_check | src/io/production_program.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/production_program.rs:100 | impactos na integridade dos dados de diagnostico
| build_conformance_summary | src/io/production_program.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/production_program.rs:100 | impactos na integridade dos dados de diagnostico
| run_uds_runtime_self_test | src/io/production_program.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/production_program.rs:100 | impactos na integridade dos dados de diagnostico
| supports | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/vendor_cat_comm.rs:100 | impactos na integridade dos dados de diagnostico
| open | src/io/vendor_cat_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs ; src/io/hw.rs ; src/io/hw.rs ; src/io/replay.rs | src/io/hw.rs:100 ; src/io/hw.rs:101 ; src/io/hw.rs:102 ; src/io/hw.rs:103 ; src/io/replay.rs:100 | impactos na integridade dos dados de diagnostico
| ensure_connected | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/vendor_cat_comm.rs:100 | impactos na integridade dos dados de diagnostico
| send_bridge_request | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/vendor_cat_comm.rs:100 | impactos na integridade dos dados de diagnostico
| read_bridge_response | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/vendor_cat_comm.rs:100 | impactos na integridade dos dados de diagnostico
| append_diag_record | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/vendor_cat_comm.rs:100 | impactos na integridade dos dados de diagnostico
| init | src/io/vendor_cat_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs ; src/io/live_runner.rs ; src/io/live_runner.rs | src/io/hw.rs:100 ; src/io/hw.rs:101 ; src/io/live_runner.rs:100 ; src/io/live_runner.rs:101 | impactos na integridade dos dados de diagnostico
| read_frame | src/io/vendor_cat_comm.rs | critica | src/io/hw.rs ; src/io/live_runner.rs ; src/io/live_runner.rs ; src/io/live_runner.rs | src/io/hw.rs:100 ; src/io/live_runner.rs:100 ; src/io/live_runner.rs:101 ; src/io/live_runner.rs:102 | impactos na integridade dos dados de diagnostico
| try_read_frame | src/io/vendor_cat_comm.rs | critica | src/io/hw.rs ; src/io/mock.rs ; src/io/serial_adapter.rs ; src/io/vendor_cat_comm.rs | src/io/hw.rs:100 ; src/io/mock.rs:100 ; src/io/serial_adapter.rs:100 ; src/io/vendor_cat_comm.rs:100 | impactos na integridade dos dados de diagnostico
| send_frame | src/io/vendor_cat_comm.rs | critica | src/io/hw.rs ; src/io/live_runner.rs ; src/io/live_runner.rs ; src/io/live_runner.rs ; src/io/live_runner.rs | src/io/hw.rs:100 ; src/io/live_runner.rs:100 ; src/io/live_runner.rs:101 ; src/io/live_runner.rs:102 ; src/io/live_runner.rs:103 | impactos na integridade dos dados de diagnostico
| close | src/io/vendor_cat_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs ; src/io/live_runner.rs ; src/io/live_runner.rs | src/io/hw.rs:100 ; src/io/hw.rs:101 ; src/io/live_runner.rs:100 ; src/io/live_runner.rs:101 | impactos na integridade dos dados de diagnostico
| adapter_info | src/io/vendor_cat_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs | src/io/vendor_cat_comm.rs:305 ; src/io/vendor_cat_comm.rs:306 | impactos na integridade dos dados de diagnostico

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| resolve_bridge_exe | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/

| map_bridge_error | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/

| bridge_info_from_response | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura estat

| ensure_bridge_compatibility | src/io/vendor_cat_comm.rs | critica | chamadas diretas nao detectadas por varredura esta

| now_ms | src/io/vendor_cat_comm.rs | critica | src/bin/cat_comm_bridge.rs ; src/bin/cat_comm_bridge.rs ; src/io/hw.rs |

| engineering_oil_catalog | src/leak_physics.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/

| uds_process_sa | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.rs:6236 |

| uds_send_and_log | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.rs:6246 |

| run_uds_flash_pipeline | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.rs

| tab_uds | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.rs:9040 | impacto

| fmt | src/uds.rs | critica | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous

| new | src/uds.rs | critica | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous

| tick | src/uds.rs | critica | src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/aut

| process | src/uds.rs | critica | src/can_gateway.rs ; src/io/production_program.rs ; src/io/production_program.rs ; s

| svc_session_control | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:320

| svc_ecu_reset | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:353 | impac

| svc_clear_dtc | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:395 | impac

| svc_read_dtc | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:428 | impac

| svc_read_data_by_id | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:504

| svc_security_access | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:554

| svc_write_data_by_id | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:640

| svc_routine_control | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:684

| svc_request_download | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:733

| svc_transfer_data | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:767 |

| svc_request_transfer_exit | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs

| svc_tester_present | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:830 |

| record_dtc_set | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:850 | impac

| nrc | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:879 | impacto local

| log | src/uds.rs | critica | src/can_gateway.rs ; src/can_gateway.rs ; src/can_gateway.rs ; src/nvm.rs | src/uds.rs:8

| service_name | src/uds.rs | critica | src/main.rs | src/uds.rs:905 ; src/main.rs:6252 | pode quebrar protocolo, gate

| dtc_to_3bytes | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:926 | impac

| session_default_relocks_security | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src

| transfer_exit_rejects_incomplete_download | src/uds.rs | critica | chamadas diretas nao detectadas por varredura esta

| bin_path | tests/vendor_bridge_e2e.rs | critica | chamadas diretas nao detectadas por varredura estatica | tests/vende

| unique_temp_dir | tests/vendor_bridge_e2e.rs | critica | chamadas diretas nao detectadas por varredura estatica | tes

| cat_comm_bridge_probe_and_live_phases_flash_e2e | tests/vendor_bridge_e2e.rs | critica | chamadas diretas nao detecta

| from_path | src/io/allowlist.rs | alta | src/io/mock.rs ; src/leak_physics.rs | src/io/allowlist.rs:33 ; src/io/mock.

| is_allowed | src/io/allowlist.rs | alta | src/io/mock.rs | src/io/allowlist.rs:42 ; src/io/mock.rs:133 | pode quebrar

| per_rule_rate_limit | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/al

| validate | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:

| rule_matches_frame | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allo

| parse_u32_hex_or_dec | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/a

| parse_mask | src/io/allowlist.rs | alta | tests/io_mock.rs | src/io/allowlist.rs:146 ; tests/io_mock.rs:190 | pode qu

| match_hex_pattern | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allow

| can_rule_matches_mask | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/a

| serial_rule_matches_pattern | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | s

| write_intent_enabled | src/io/hw.rs | alta | src/io/mock.rs ; tests/io_mock.rs | src/io/hw.rs:223 ; src/io/mock.rs:12

| connect_ecm | src/io/live_runner.rs | alta | src/io/production_program.rs ; src/main.rs | src/io/live_runner.rs:174 ;

| live_preflight_ecm | src/io/live_runner.rs | alta | src/io/production_program.rs | src/io/live_runner.rs:793 ; src/io.

| check_live_channel_policy | src/io/live_runner.rs | alta | chamadas diretas nao detectadas por varredura estatica | s

| on_rate_limited | src/io/metrics.rs | alta | src/io/mock.rs | src/io/metrics.rs:37 ; src/io/mock.rs:149 | pode quebrar

| inject_disconnect | src/io/mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/mock.rs:3

| clear_disconnect | src/io/mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/mock.rs:36

| read_frame | src/io/mock.rs | alta | src/io/hw.rs ; src/io/live_runner.rs ; src/io/live_runner.rs ; src/io/live_runner

| try_read_frame | src/io/mock.rs | alta | src/io/hw.rs ; src/io/serial_adapter.rs ; src/io/socketcan_adapter.rs ; src/

| send_frame | src/io/mock.rs | alta | src/io/hw.rs ; src/io/live_runner.rs ; src/io/live_runner.rs ; src/io/serial_adap

| default | src/io/production_program.rs | alta | src/autonomous.rs ; src/bin/cat_comm_bridge.rs ; src/bin/cat_comm_bri

| run_sim_baseline_check | src/io/production_program.rs | alta | chamadas diretas nao detectadas por varredura estatica

| run_sim_connectivity_check | src/io/production_program.rs | alta | chamadas diretas nao detectadas por varredura esta

| run_sim_preflight_check | src/io/production_program.rs | alta | chamadas diretas nao detectadas por varredura estatica

| run_sim_stress_check | src/io/production_program.rs | alta | chamadas diretas nao detectadas por varredura estatica |

| hex_sha | src/io/production_program.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/produ

| now_ms | src/io/production_program.rs | alta | src/bin/cat_comm_bridge.rs ; src/bin/cat_comm_bridge.rs ; src/io/hw.rs

| new | src/io/rate_limiter.rs | alta | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/a

| try_take | src/io/rate_limiter.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/rate_limit

| refill | src/io/rate_limiter.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/rate_limiter

| new | src/io/rate_limiter.rs | alta | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/a

| check_can | src/io/rate_limiter.rs | alta | src/io/mock.rs | src/io/rate_limiter.rs:57 ; src/io/mock.rs:141 | pode qu

| check_serial | src/io/rate_limiter.rs | alta | src/io/mock.rs | src/io/rate_limiter.rs:70 ; src/io/mock.rs:142 | pode

| limiter_blocks_when_empty | src/io/rate_limiter.rs | alta | chamadas diretas nao detectadas por varredura estatica | s

| aggregate_group_report | src/leak_physics.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/lea

| export_calibration_report_json | src/leak_physics.rs | alta | src/lib.rs | src/leak_physics.rs:1654 ; src/lib.rs:1123

| export_calibration_report_csv | src/leak_physics.rs | alta | src/lib.rs | src/leak_physics.rs:1670 ; src/lib.rs:1131

| report_captures_pressure_band_and_safe_window | src/leak_physics.rs | alta | chamadas diretas nao detectadas por varre

| calibration_mode_ingests_csv_and_exports_reports | src/leak_physics.rs | alta | chamadas diretas nao detectadas por v

| export_leak_report_json | src/lib.rs | alta | src/main.rs ; tests/leak_system_integration.rs | src/lib.rs:1053 ; src/

```
| export_leak_report_csv | src/lib.rs | alta | src/main.rs ; tests/leak_system_integration.rs | src/lib.rs:1060 ; src/main.rs:1060
| export_leak_calibration_report_json | src/lib.rs | alta | src/main.rs | src/lib.rs:1118 ; src/main.rs:1142 | pode quebrar
| export_leak_calibration_report_csv | src/lib.rs | alta | src/main.rs | src/lib.rs:1126 ; src/main.rs:1130 | pode quebrar
| pick_save_path | src/main.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/main.rs:542 | impacto
| leak_asciicad | src/main.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/main.rs:6360 | impacto
| fmt | src/v2x_telematics.rs | alta | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs
| write_test_allowlist | tests/io_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | tests/io_mock.rs:100
| mock_adapter_allows_write_when_policy_satisfied | tests/io_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica
| mock_adapter_dry_run_logs_without_physical_tx | tests/io_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica
| mock_adapter_rate_limit_blocks_burst | tests/io_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica
| allowlist_parse_mask_accepts_hex | tests/io_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica
| e2e_hydraulic_plus_ac_stress_generates_reports | tests/leak_system_integration.rs | alta | chamadas diretas nao detectadas
```

</details>

3. Roadmap de Refatoracao Segura por Ondas

Wave 1 - Blindagem de Contratos de Protocolo

- Escopo: src/io/live_runner.rs, src/io/vendor_cat_comm.rs, src/uds.rs
- Mudancas: centralizar validacao de SID positivo, tipar NRC, consolidar parse de DIDs
- Checklist de testes:
- [] cargo check
- [] cargo test --workspace
- [] cargo test --features vendor-windows --test vendor_bridge_e2e

Wave 2 - Refino de Abstracao de Hardware

- Escopo: src/io/hw.rs, src/io/socketcan_adapter.rs, src/io/serial_adapter.rs
- Mudancas: contratos de capability com traits menores, erros padronizados por camada
- Checklist de testes:
- [] cargo check --features vendor-windows
- [] cargo test --workspace
- [] cargo test --test io_mock

Wave 3 - Estabilizacao de Schema de Relatorio

- Escopo: src/io/production_program.rs, src/bin/simulator_cli.rs, docs/ECM-Data.md
- Mudancas: versionamento de schema, compat adapters para consumidores antigos
- Checklist de testes:
- [] cargo check
- [] cargo test --workspace
- [] powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1

Wave 4 - Performance e Observabilidade

- Escopo: src/main.rs, src/observability.rs, src/io/metrics.rs
- Mudancas: reduzir jitter de render e padronizar eventos de telemetria
- Checklist de testes:

- [] cargo check
- [] cargo test --workspace

Wave 5 - Harden de CI e Evidencia

- Escopo: .github/workflows/*.yml, scripts/*.ps1, scripts/*.sh
- Mudancas: gates obrigatorios por risco, artefatos de report/hash sempre anexados
- Checklist de testes:
- [] cargo check
- [] cargo test --workspace
- [] cargo test --features vendor-windows --test vendor_bridge_e2e

Cobertura e Limites

- Esta edicao prioriza funcoes de alto impacto e rastreabilidade de chamadas por varredura estatica.
- Chamadas dinamicas e dispatch indireto podem nao aparecer na tabela X->Y.